Homework 4: SOLID Principles

CS 150 with Dr. Sam Schwartz

Upload to Canvas when done.

1 Purpose

In this assignment you will gain experience with SOLID principles.

2 Tasks

This will be a written homework assignment. There are five SOLID principles:

1. Single responsibility
2. Open-closed
3. Liskov substitution
4. Interface segregation
5. Dependency inversion

For each principle listed above, do the following:

1. Define the principle using professional English. (5pts)
2. Give a code example of the principle being correctly used. (5pts)
3. Give a "smelly" code example where the principle is not followed (5pts)
4. ... with a brief explanation written in professional English of why your flawed example code "smells" when the principle is violated. (5pts)

Submit to Canvas

Upload the following files to Canvas:

- One PDF document, with your name at the top. Each principle should begin on its own page, with a header stating the principle. The font should be a typical default font for prose (e.g., Ariel, Times New Roman, Styx, Calibri, etc.) with a monospaced font for code (e.g., Courier New).
- (Optional) For an extra 10 points, you may also submit a single `.tex` file which corresponds to the PDF you uploaded. All computing students (and, frankly, all college students) should learn how to use LaTeX sooner rather than later. LaTeX (pronounced either Lah-Tech or Lay-Tech; both are acceptable although some people find one or the other to be "correct") is the standard document creation software used in computer science and mathematics. It is how you can type expressions such as

  $$\sum_{\alpha=1}^{\Omega} \left(\int_{y}^{x} \alpha^{\beta} \quad \mathrm{d}\beta \right)$$

  very quickly. It's also how you can insert code snippets into a document professionally.

  I write nearly everything (documents, letters, this homework assignment) in LaTeX, to the point that I don't even have MS Word installed on my computer. There is a steep learning curve at first. But if you struggle through the steep learning curve, you will save hundreds of hours of time later on. Hence, I'm offering you some extra credit if you use LaTeX for this assignment.